

Automatic Audio Augmentation for Requests Sub-Challenge

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ABSTRACT

This paper presents our solution for the Requests Sub-challenge of the ACM Multimedia 2023 Computational Paralinguistics Challenge. Drawing upon the framework of self-supervised learning, we put forth an automated data augmentation technique for audio classification, accompanied by a multi-channel fusion strategy aimed at enhancing overall performance. Specifically, to tackle the issue of imbalanced classes in complaint classification, we propose an audio data augmentation method that generates appropriate augmentation strategies for the challenge dataset. Furthermore, recognizing the distinctive characteristics of the dual-channel HC-C dataset, we individually evaluate the classification performance of the left channel, right channel, channel difference, and channel sum, subsequently selecting the optimal integration approach. Our approach yields a significant improvement in performance when compared to the competitive baselines, particularly in the context of the complaint task. Moreover, our method demonstrates noteworthy cross-task transferability.

CCS CONCEPTS

• **Computing methodologies** → *Supervised learning by classification; Knowledge representation and reasoning.*

KEYWORDS

Computational Paralinguistics, Automatic Audio Augmentation, Data Augmentation, Audio Classification

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1 INTRODUCTION

This year's ACM Multimedia Computational Paralinguistics Challenge introduced two intriguing sub-challenges [17]. This paper addresses the Requests Sub-challenge by proposing an automatic audio data augmentation method and fully leveraging the dual-channel information. The Requests Sub-challenge encompasses two binary classification tasks: Complaint and Request. In the call center (CC) scenario, having a high-performance classifier is crucial for effective customer relationship management (CRM) [2, 8, 16].

Data augmentation is a highly effective approach for enhancing the generalization capability of binary classification models in real-world scenarios and addressing challenges posed by class imbalance. In a recent study, Iqbal et al. [10] conducted a comprehensive review of audio data augmentation techniques within the context of deep learning. They classified audio feature augmentation strategies into two main types: waveform-based and spectrogram-based methods. Waveform-based augmentations commonly include time stretching, pitch shifting, speed perturbation, and the addition of background noise [1, 15]. Another approach involves applying image augmentation operations to spectrograms [11, 13, 14, 20, 22, 27]. However, there is currently no universally applicable data augmentation strategy that encompasses both waveform and spectrogram features. Furthermore, the selection of suitable augmentation strategies and determination of their parameters often requires significant manual effort. For instance, experts must manually design the masking ratio in SpecAugment [14] and the mixing ratio in Mixup [13], tailoring them to specific tasks.

This paper presents the Automatic Audio Augmentation method as a means of generating augmentation strategies for the challenge datasets. The primary goals of this approach include minimizing the need for manual design, improving the generalization performance of models, and addressing class imbalance issues. Furthermore, a dual-channel audio fusion strategy is introduced, specifically tailored to the unique characteristics of dual-channel HC-C data.

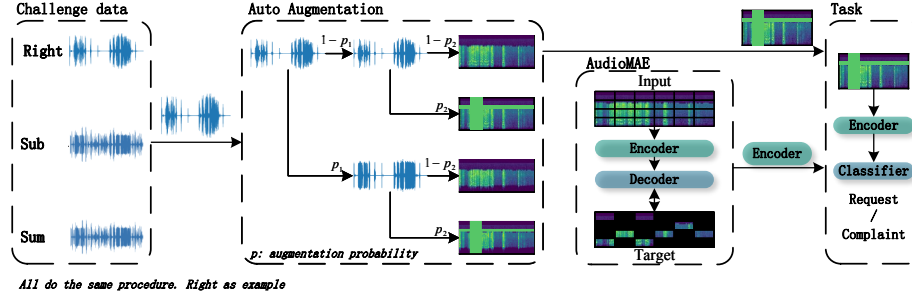


Figure 1: The framework of the proposed approach.

2 METHODOLOGY

The diagram presented in Figure 1 depicts the comprehensive pipeline of the solution proposed in this research. Our methodology comprises three primary elements: challenge data preprocessing, automatic audio augmentation design, and classification. It is noteworthy that the AudioMAE model serves as the fundamental backbone in our experiments. Subsequently, the subsequent section presents an elaborate elucidation of the architecture of each component within our framework.

2.1 Challenge Data Preprocess

The HC-C dataset was used in the Requests sub-challenge, including dual-channel audio recordings. Each audio segment consisted of two channels: wav_{left} and wav_{right} . The wav_{left} channel contained the customer's audio, while the wav_{right} channel contained the agent's audio. Two models, f^{left} and f^{right} , were trained on these channels separately. For further analysis, two dual-channel models were proposed: one leveraging the channel difference wav_{sub} , and the other utilizing the channel sum wav_{sum} . f^{sub} was developed using wav_{sub} as input to capture the differences between the left and right channels. f^{sum} was developed by using wav_{sum} as input to emphasize the overall mixed audio.

$$wav_{sub} = \frac{wav_{left} - wav_{right}}{2}, \quad (1)$$

$$wav_{sum} = \frac{wav_{left} + wav_{right}}{2}. \quad (2)$$

However, the performance analysis of the local-global channel models on the validation set showed unsatisfactory results for the model based on the customer's audio (wav_{left}). As a result, the model f^{left} was discarded during the final fusion of the multi-channel models. The final model f^* , was obtained through arithmetic averaging.

$$f^* = \frac{f^{right} + f^{sub} + f^{sum}}{3} \quad (3)$$

2.2 Automatic Audio Augmentation

In order to augment the diversity of training samples and address the issue of class imbalance, we present an innovative data augmentation methodology that incorporates both waveform-level and spectrogram-level augmentations. This methodology integrates

various augmentation operations at different stages, thereby facilitating a more comprehensive transformation. Our augmentation strategy comprises three distinct sub-strategies: waveform augmentation, denoted as O^{wave} , spectrogram augmentation, denoted as O^{spec} , and mix augmentation, denoted as O^{mix} . We shall henceforth refer to this approach as augmentation strategy $\mathcal{P}$, which can be mathematically expressed as follows:

$$\mathcal{P} = \{O^{wave}, O^{spec}, O^{mix}\}. \quad (4)$$

We introduce the symbol t as a representation of the augmentation operation, where t can denote a specific transformation such as time shifting or frequency masking. The variables p and m are used to denote the probability and magnitude, respectively, of applying the operation t , with p and m both constrained within the interval $[0, 1]$. In our notation, an audio sample is represented by the variable x , which can correspond to various channels such as the left channel, right channel, channel sum, or channel difference. We denote the resulting augmented audio sample as x_a , which can be formally expressed as follows:

$$x_a = O_k^{mix} \{O_j^{spec} [O_i^{wave}(x; \langle t_i, p_i \rangle); \langle t_j, p_j, m_j \rangle]; \langle t_k, p_k \rangle\}. \quad (5)$$

Our augmentation strategy combines operation t , execution probability p , and execution magnitude m to control the augmentation range. In this challenge, we used 18 operations and 21 execution parameters. Designing these parameters manually is challenging, and brute force search is inefficient. Therefore, an algorithm for efficient automatic audio augmentation is proposed. The algorithm consists of two steps: (1) generating candidate augmentation strategies and (2) selecting appropriate strategies. First, referring to the work in [12], a set of candidate strategies $\mathcal{B}$ is generated using variable kernel density estimation, as shown in Equation 6. We use TPE optimization methods with HyperOpt [4] for efficient implementation.

$$\mathcal{B} = \{\mathcal{P}_1(O^{wave}, O^{spec}, O^{mix}), \dots, \mathcal{P}_B(O^{wave}, O^{spec}, O^{mix})\}. \quad (6)$$

Now there is a candidate strategy set $\mathcal{B}$, and the next step is to determine the strategy selection function. Unlike [12] in selecting the top N strategies in $\mathcal{B}$, we introduce the accuracy matching coefficient λ to select a more robust strategy set. The goal is to choose strategies close to the target accuracy. Equation 7 displays the strategy selection function, where $\mathcal{A}$ represents the target augmentation strategy set.

$$\mathcal{A} = \{\mathcal{P} \mid \left| \frac{\mathcal{H}(f^*|D_{sample}(\mathcal{P}))}{\mathcal{H}(f^*|D_{val}(\mathcal{P}))} - 1 \right| \leq \lambda, \mathcal{P} \in \mathcal{B}\}, \quad (7)$$

2.3 AudioMAE

With the proliferation of audio data, numerous studies have explored using self-supervised learning techniques to pretrain neural network models using unlabeled data, followed by fine-tuning with labeled data for improved performance in classification tasks [21, 25, 28]. Several noteworthy techniques have emerged in the field of audio classification self-supervised learning, including SSAST [7], AMT [23, 24, 26], and AudioMAE [9], among others. The network architecture employed in this study draws inspiration from AudioMAE, which is based on the Masked Autoencoder (MAE) methodology. Similar approaches encompass MAE-AST [3] and MaskSpec [5]. Notably, AudioMAE has demonstrated exceptional performance across multiple datasets. Consequently, our work builds upon the model of AudioMAE, which is pretrained on the AudioSet dataset [6]. Our work is innovating algorithms for the fine-tuning stage and finally serving for the Requests sub-challenge.

3 EXPERIMENTS AND RESULTS

Based on the challenge dataset, this paper conducts comprehensive experiments to validate the performance of the final model. Additionally, we perform ablation experiments to assess the effectiveness of our methods and subsequently discuss the transferability of the proposed automatic audio augmentation strategy.

3.1 Quantitative results

Request: The optimal UAR value of 64.02% is achieved on the Test dataset, as presented in Table 1. By referring to Figure 2a, which corresponds to the superior Dev outcome indicated in Table 1, it can be observed that the modeling of “presta” outperforms that of “affil”.

Complaint: We obtain the best **UAR=54.19%** on Test, see Table 1. Figure 2b shows, for the best Dev result given in Table 1, that “no” can be better modelled than “yes”, with a great improvement.

The overall result for the Requests Sub-Challenge is the highest UAR obtained from both request and complaint: $(64.02 + 54.19) / 2 = 59.11\%$. It is important to note that our primary contribution lies in the proposed plug-and-play AAA method, exemplified in Table 2. The overall performance improvement is calculated as $(4.29 + 3.45) / 2 = 3.87\%$, compared without augmentations.

3.2 Ablation Studies

3.2.1 Selection of Multi-channel Representations. This part examines the classification performance of different channel representations on the dual-channel audio dataset utilized in this challenge. Our experiments involve analyzing the left channel, right channel, channel difference, and channel sum. The results, presented in Table 3, reveal unexpected findings. Surprisingly, the classification performance of the model based solely on the left channel (which captures the customer’s audio in channel 0) is inadequate. Consequently, we made the decision to exclude the left channel representation and solely incorporate models based on the right channel, channel

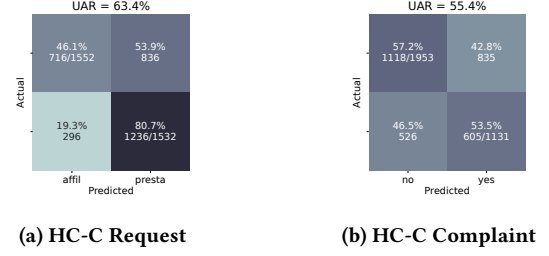


Figure 2: Confusion matrices for (a) HC-C Request and (b) HC-C Complaint.

difference, and channel sum for audio representation in our final channel fusion.

3.2.2 Improvement of Automatic Audio Augmentation Method. As aforementioned, AAA aims to address resource scarcity and class imbalance problems with minimal human intervention. In this part, we evaluate the performance improvement of AAA in challenging contexts using the HC-C dataset. Table 2 presents the classification performance of three variants of AudioMAE on four different channel models. The “no-aug” control group underwent no augmentation, while the “ori-aug” group used a common initial augmentation strategy. The “auto-aug” group employed the AAA method, significantly improving classification performance on both tasks across all channels. The “improvement” metric represents gains achieved by the AAA method on multiple channels and overall performance improvement on both tasks. The HC-C dataset shows a significant data distribution imbalance in the Complaint task, with a Yes:No ratio of approximately 1:2.2 in the training and validation sets. This class distribution imbalance presents challenges for the classification task. By employing various augmentation strategies, the AAA method enhances sample diversity, expands the dataset size, balances extreme categories, and enhances the model’s generalization ability. Therefore, the AAA method provides a viable solution for addressing class imbalance challenges and achieves superior performance in the class-imbalanced Complaint task compared to the baseline.

We analyzed the augmentation strategies for the two subtasks and visually represented the results using stacked bar graphs, as depicted in Figure 3. The strategies for each dataset were categorized into three groups based on sub-strategy types: waveform level, spectrogram level, and mixup level, listed in descending order. Longer bars in Figure 3 indicate a higher frequency of usage. Our analysis of the bar graphs revealed that the frequency of operations under each sub-strategy is relatively balanced, with each operation being selected. Furthermore, the visualization demonstrated that the usage probability distribution of the two subtasks within the same task is similar. This characteristic will be further examined in the next section through transferability analysis.

3.3 Transferability

Figure 4 reveals similarities in the distribution of policy parameters across different tasks. Therefore, this section evaluates the transferability of the augmentation policy generated by the AAA method across different tasks within the same dataset. It is worth

Table 1: Results for the Sub-Challenges HC-C. CI on Test: Confidence Intervals on Test. *: Confidence Intervals on Dev.

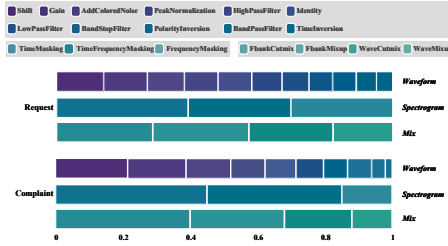
[UAR %] Approach	HC-C Request			HC-C Complaint		
	Dev	Test	CI on Test	Dev	Test	CI on Test
wav2vec	65.1	67.2	65.7-68.7	50.9	52.2	50.7-53.7
auDeep	54.9	52.4	50.7-54.0	50.9	50.9	49.0-52.7
DeepSpectrum	58.2	55.6	54.0-57.2	49.6	51.7	50.1-53.4
ComParE	60.9	58.8	57.2-60.5	<u>51.8</u>	<u>52.9</u>	51.2-54.7
Late Fusion	60.5	59.5	57.8-61.2	51.2	51.8	50.1-53.5
Our	<u>63.41</u>	<u>64.02</u>	61.86 – 64.98*	55.37	54.19	53.58 – 57.14*

Table 2: The improvement of Automatic Audio Augment in Sub-Challenge HC-C on Dev.

[UAR %] Channel	Request				Complaint			
	no-aug	ori-aug	auto-aug	improvement	no-aug	ori-aug	auto-aug	improvement
Right	<u>58.97</u>	59.24	62.00	3.03	51.80	<u>51.69</u>	53.90	1.21
Sub	58.68	<u>57.91</u>	62.02	4.11	<u>50.30</u>	50.78	54.29	2.99
Sum	<u>58.99</u>	59.50	61.80	2.81	<u>50.72</u>	50.82	52.81	2.09
Our	<u>59.19</u>	59.99	63.48	4.29	<u>51.92</u>	52.10	55.37	3.45

Table 3: Results for the multi-channel representations.

[UAR %] Channel	HC-C Request		HC-C Complaint	
	Dev	CI on Dev	Dev	CI on Dev
left	55.67	54.05-57.23	52.08	50.25-53.90
right	62.00	60.32-63.71	53.90	52.01-55.59
sub	62.02	60.39-63.62	54.29	52.59-55.95
sum	61.80	60.10-63.52	53.81	51.99-55.65

**Figure 3: The final policy distribution graph of HC-C records the execution frequency and probability of each operation. The length of each bar represents the usage frequency.**

mentioning that both tasks utilize the HC-C audio dataset, sharing training and validation audio samples but differ in the task labels. We examine the cross-task transferability of the generated policies to assess any performance degradation that may occur when applying augmentation policies from source tasks to target tasks.

As depicted in Figure 4, the augmentation policies generated for the complaint subtask resulted in a mere 0.33% performance drop when applied to the request subtask. Similarly, the augmentation policies generated for the request subtask led to a slight 0.28% performance drop when applied to the complaint subtask. Considering the significant performance improvements of 3.45% and 4.29% achieved by the AAA method on the two tasks, these slight performance drops are deemed acceptable. Consequently, we

can conclude that the data augmentation strategy employed by the AAA method demonstrates transferability across different tasks.

**Figure 4: Transferability experiment of the AAA method from source task to target task. The number presents the UAR(%) degradation when the policy is transferred.**

4 CONCLUSION

We used Automatic Audio Augmentation for the Requests sub-challenge in the ACM Multimedia Computational Paralinguistics Challenge. Our algorithm employs self-supervised learning to generate diverse augmentation strategies tailored to the HC-C dataset. We propose a multi-channel training fusion strategy to accommodate the dual-channel characteristics of the dataset. Our method significantly improves performance, especially in the Complaint task, and demonstrates the potential for cross-task transferability.

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